Principal Component Analysis (PCA) & t-distributed Stochastic Neighbor Embedding (t-SNE)

Intelligent Systems

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Principal Component Analysis

Principal Component Analysis

Principal Component Analysis, or **PCA**, is a dimensionality reduction machine learning method, used to simplify a large data set into a smaller set of **principal components**, which capture the most variability with minimal loss of information:

→ Reduces data complexity

→ Reduces noise and improves interpretability

→ Enhances visualization in fewer dimensions.

This technique can be used for:

→ Image compression

→ **Feature extraction** in *Machine Learning*

→ Genomic data analysis.

Principal Component Analysis

PCA can be broken down into five steps:

i) Standardize the data → Normalize variables to ensure same scale

ii) Compute the **covariance matrix** → Identify correlations

iii) Calculate **eigenvectors** and **eigenvalues** → Eigenvectors define principal component directions; eigenvalues measure their importance

iv) Select principal components → Choose the most significant components based on eigenvalues

v) Recast the data → Project the original data onto the new principal component axes.

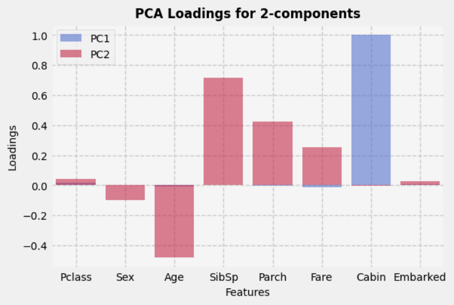

Figure 1: PCA for Titanic dataset

t-distributed Stochastic Neighbor Embedding

t-distributed Stochastic Neighbor Embedding

t-distributed Stochastic Neighbor Embedding, or ***t*-SNE**, is a non-linear dimensionality reduction technique primarily used for **visualizing high-dimensional data** in a lower-dimensional space (typically 2D or 3D). Unlike **PCA**, which focuses on global structure, t-SNE preserves local relationships between data points:

- → Captures non-linear relationships
- → Preserves local structures in data
- → Enhances cluster visualization.

This technique is commonly used for:

- → High-dimensional data visualization
- → **Clustering analysis** in *Machine Learning*
- → Exploring word embeddings and genomic data.

t-distributed Stochastic Neighbor Embedding

***t*-SNE** operates in the following steps:

i) Compute **pairwise similarities** in high-dimensional space → Use *Gaussian* probability distributions

ii) Map data points to lower-dimensional space → Convert similarities into *Student's* t-distributions

iii) Optimize positioning → Use **gradient descent** to minimize *Kullback-Leibler* divergence

iv) Preserve local structures → Nearby points in high-dimensional space remain close in low-dimensional space.

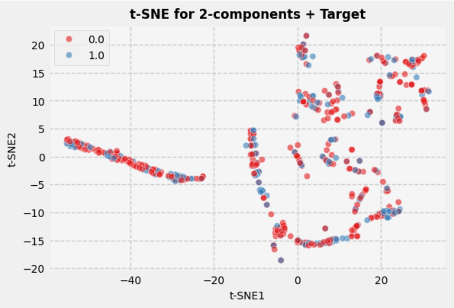

Figure 2: t-SNE for Titanic dataset

References

References i

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